

fast. To install JACK, download the latest source code (jack-audio-connection-kit-0.80.0.tar.gz) from <http://jackit.sourceforge.net>, and then uncompress and compile using the standard commands.

After this, you'll have to install the LADSPA plugins that allow for easy support of third-party plugins.

Linux Audio Developer's Simple Plug-in API (LADSPA): LADSPA is a simple and flexible API that allows programmers to develop new audio processing programs that can be easily plugged into many existing LADSPA-compliant software. This eliminates the problem of developers coming up with several different applications that use even more diverse libraries to serve their purpose. This leads to a very complex situation where you have to download and install not just the application, but a whole set of libraries for each software you try. LADSPA provides a unified interface that allows any LADSPA compliant plug-in to work with any other LADSPA compliant application.

To install LADSPA, first download the LADSPA SDK (ladspa_sdk.tgz) from www.ladspa.org/download/. Next, uncompress the package using the `'tar xzvf ladspa_sdk.tgz'` command.

After this, `'cd'` to the directory `'ladspa_sdk/src'` and type `'make'` to compile the source code.

The compilation will take a few minutes, after which you can test if it was successful by using the following commands:

```
'play ../snd/noise.wav'
```

```
'play /tmp/test.wav'
```

If the test is successful, use the `'make install'` command to install the LADSPA libraries in their respective directories. The

LADSPA SDK currently comes with a basic set of plug-ins, but more advanced plug-ins are also available—visit the LADSPA Web site for links to these sites.

Now that you've set up all the drivers and plug-ins, let's install the actual audio editing software. From the plethora of audio editing software currently available, we chose to install Audacity, since it is both easy to install and ideal for novices.

Audacity

Audacity is a free audio editor that is available on multiple platforms including Windows, Linux, Unix and Mac OS X. It can record as well as play sounds, import



Audacity has been ported to many operating systems including Mac OS X

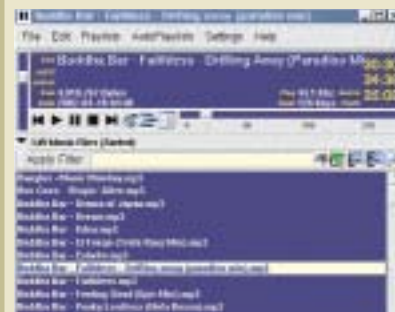
and export WAV, AIFF, and MP3 files, and more. Its simple-to-use interface lets you edit your sounds—using simple Cut, Copy and Paste operations, with unlimited Undo—as well as mix tracks, or apply effects to your existing recordings. It also has a built-in amplitude envelope editor, a customisable spectrogram mode and a frequency analysis window for audio

filters, mixers, faders, etc., that can be used to build your own audio sub-system. The audio driver is just one of the main components that interfaces with the sound card and plays data via the speakers.

GStreamer: GStreamer allows for the construction of graphs of media-handling components, ranging from simple MP3 playback to complex audio (mixing) and video (non-linear editing) processing. Applications can take advantage of advances in codec and filter technology transparently. Since GStreamer is released under the LGPL, developers can add new codecs and filters by writing a simple plug-in with a clean, generic interface.

The Replacement Player

SnackAmp: SnackAmp is a less popular but powerful music player on Linux. Its interface is more complex and rudimenta-



SnackAmp isn't as basic as it looks; it has some really cool features

ry than XMMS, but once you master its interface you get control over some amazing features that are absent in XMMS, such as cataloguing and managing folders and removable media.

analysis applications. Apart from built-in effects such as Bass Boost, Wahwah, and Noise Removal, it has support for VST plug-ins that are popular on Windows.

Installing Audacity is very simple, provided you have all the required libraries. While the Ogg and Vorbis libraries generally come pre-installed with any Linux distribution, Audacity requires a special GUI component (called wxGTK) that is required to generate its interface. This can be downloaded as an rpm file from the audacity Web site and installed using the `'rpm -ivh <filename>'` command. Once the required libraries are installed, just use the `'tar'`, `'./configure'`, `'make'`, and `'make install'` commands (for source code), or the `'rpm'` command (for binary code) to install Audacity.

Conclusion

Overall, Linux is a very viable option, not just for setting up an MP3 jukebox, but also for more advanced work such as audio editing workstations, or even recording studios. The important factor is the effort required in setting up all these drivers and applications, and then optimising them to get the very best out of the hardware. With specialised, sophisticated drivers such as JACK and LADSPA becoming more stable and easy to configure, it seems likely that Linux will soon make waves in this field. ■

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Backseat Drivers

Open Sound System (OSS): This is the Linux sound driver from 4Front Technologies (the makers of XMMS). While these are the default drivers that ship with Linux distributions, they are being slowly replaced by the far more advanced ALSA drivers.

Enlightenment Sound Daemon (ESD): ESD is an audio driver that has its roots in the Enlightenment window manager project. Although initially meant for the Enlightenment system, it has detached itself from the window manager, and is now available as separate drivers irrespective of the GUI you choose to run on your Linux system.

Analog Realtime Synthesizer (aRts): aRts has numerous modules such as oscillators,